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Reya: Reuse Yarn. Product, Process and Practice

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Reya: Reuse Yarn. Product, Process and Practice

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Reya: Reuse Yarn is a practice-led research project that investigates scalable methods for reducing pre-consumer waste in commercial knitwear manufacturing. Positioned within a circular manufacturing framework, the study re-evaluates knit as product, process, and practice to generate new insights into sustainable production. The research addresses the environmental and economic impacts of deadstock yarn, particularly low-value synthetic blends that are typically discarded due to limitations in quantity, aesthetics, and material consistency. Drawing on interdisciplinary knowledge from knit and visual colour psychology, the project employs a *Knit Thinking* methodology, combining tacit material understanding with explicit technical knowledge through iterative, material-led enquiry. The research exploits optical colour mixing in knitted fabrics, where blended yarns and stitch structures create a perceptually unified colour rather than identifying individual colours in the blend of small, evenly distributed contiguous points of colour. This demonstrates minor colour inconsistency with yarn substitutions. A pilot study and two commercial trials demonstrated the viability of this approach. Findings reveal that optical mixing enables the effective use of small quantities of yarns while maintaining consistent garment aesthetics and handle. Consumer testing further indicates widespread acceptance of minor colour variation, challenging industry assumptions regarding the necessity of colour uniformity. The research concludes that integrating blended yarn techniques into existing manufacturing processes offers an immediately implementable, cost-neutral strategy for waste reduction. By reframing acceptable colour tolerance, *Reya: Reuse Yarn* proposes a shift in industry practice that supports circularity, reduces environmental impact, and aligns with emerging legislative pressures such as Extended Producer Responsibility.

Keywords: Knit Thinking; Deadstock Waste Reduction; Yarn; Colour perception.

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1. Introduction

Knit is a product, a process and a practice. This research asserts that by re-evaluating the product and process, new insights into knit practice will emerge. In this paper, the methodological positioning of Reya: Reuse Yarn, a research project that investigates new production methods in commercial knitwear manufacture, is presented through an analysis of an interdisciplinary, material-led enquiry. It utilises the embodied knowledge of knitting and explicit colour psychology knowledge to propose a new manufacturing practice that could have a positive environmental impact across the broader global manufacturing industry. Reya: Reuse yarn sits within a framework of circular manufacturing (figure 1), which must address the product – what is made, the process – how it is made, and ultimately the manufacturing practice that supports over-production and consumption.

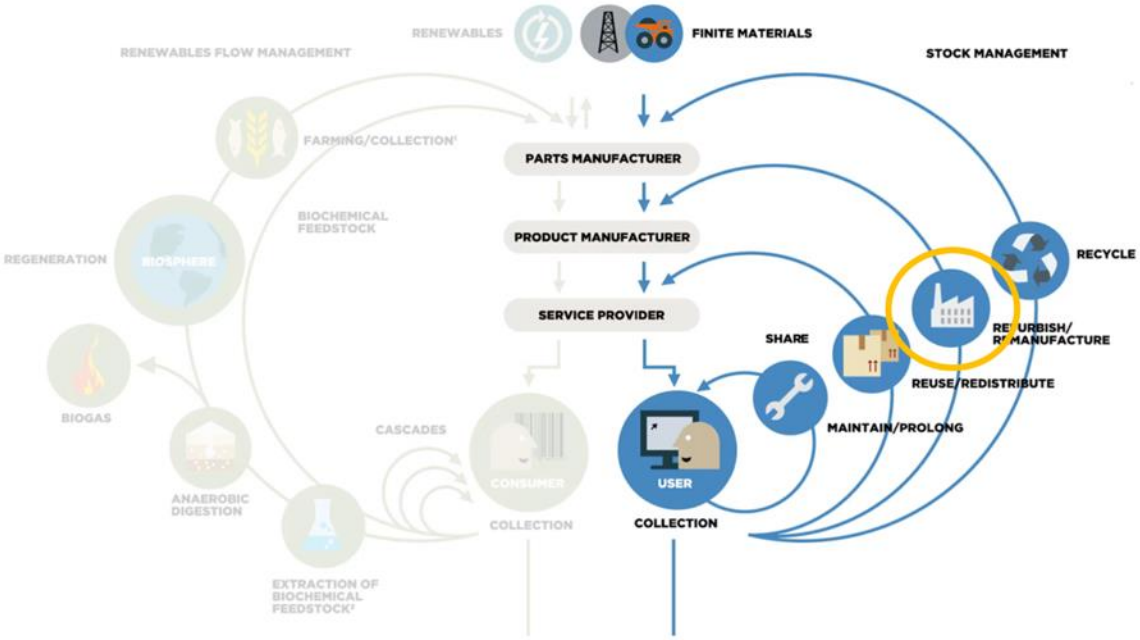


Figure 1: Technical circular manufacturing identifying the place of reuse (Ellen Macartney Foundation, 2022).

2. The Societal Problem: Waste Reduction

The reduction of the negative environmental impact of the global fashion industry is a priority across the fashion supply chain and intervention is required at each processing stage (Palm et al, 2021). Multiple strategies are evident however, 43% of UK manufacturers view waste reduction as a priority (Mansell, 2026). The current linear production model results in 1.2bn tonnes of global carbon emissions and $500bn lost to waste annually (Ellen Macarthur Foundation, 2017). Reducing waste has the dual benefits of reduced environmental and financial cost (Environment Agency, 2018) as well as fulfilling the legislative requirements of extended producer responsibility (Ellen Macarthur Foundation, 2024). Reya: Reuse Yarn addresses pre-manufacturing waste (deadstock yarn) in high-volume knitwear manufacture.

In knitwear, seasonal colour choices result in small quantities of yarn in specific colours and counts leftover at the end of the production cycle. Craft-based, hand-knitters have long used “stash-busting” patterns to utilise small quantities left over at the end of a project. These methods are replicated in

luxury knitwear production using high-value yarns such as wool, silk and cashmere, where accessories, micro-patterned garments or reprocessing use up small quantities of deadstock yarns.

Currently, there isn't a scalable route for low-value yarns such as synthetic and synthetic blend yarns, which make up the bulk of high-volume knitwear production. Carvalho et al (2025) identified that conventionally, these yarns are stored for approximately 3 years before incineration or removal to landfill. There are valid barriers to the reuse of deadstock yarn. Fashion depends on newness, and micro-seasonal design choices restrict yarn reuse on aesthetic grounds. Additionally, small quantities of yarn don't fit into volume production models. Crucially, as the fabric and garment are produced synchronously in knitwear, deviation in material use can affect not only garment aesthetics, but sizing and fabric handle.

3. The Research Context

Reya: Reuse yarn was initiated as product/process-focused research, designed to solve a tangible problem; however, the research evolved from a speculative investigation into colour, pattern and perception in knitted fabric initiated by design curiosity, rather than a pragmatic product outcome (Gaston, 2016). Through iterative design interventions (figure 2), this eventually became a study of the insubstantiality of colour (Swirnoff, 2003) in knitted fabrics within changing environments. It primarily explored optical mixing, where the brain perceives an overall colour tint, rather than identifying individual colours in a blend of small, evenly distributed contiguous points of colour (Day, 1933). In knit, the smallest unit of colour possible is the individual stitch.



Figure 2: Colour design research preceding Reya: Reuse Yarn.

The effect of optical mixing is enhanced in knitted fabric through texture (Newark, 2007). The V shape of knitted fabrics and the hairiness of yarn used produce a blurred border between colours, which

reduces the perception of colour intensity and lowers the border contrast between colours (Beck, 1972; Kanizsa, 1979). This produces ideal conditions for optical mixing.

The knowledge generated from this research was embedded in fabric; here, knowledge is construed broadly as an affect on the world (Plotkin, 1993). It was revealed by the maker through iterative practice where the repetition of action improved not only the control of the action but also developed the understanding of the action (Wilson, 1998, Ingold, 2013, Nimkulrat, 2024). The reciprocity of forming something and being changed by the process led to a transformation in thinking (Shaefer-Simmern, 1990) and a defined research question: "How can the knowledge generated from this colour research be revealed in a pragmatic design solution?"

4. Methodology

Positive change can only occur with acceptance from all stakeholders, including manufacturers, designers and consumers. New products and processes must address cognitive barriers to change, including design/production knowledge gaps and a willingness to change embedded production processes. Technical barriers, including yarn quantity and aesthetic/handle inconsistency, require solutions without incurring additional costs.

This complex research relationship was explored with a Knit Thinking methodology. Knit thinking-based research uses exploratory abductive reasoning to combine tacit craft knowledge and explicit technical knowledge reflectively and intuitively (Dudley and Mealing, 2000) in the craft-based domain of materials, process and skill (Gaston and Scott, 2021).

Knit Thinking extends the Design Thinking/Craft Thinking/Textiles Thinking canon. Detailed discussions of these models exist elsewhere; here, the key tenets relative to this paper are provided to clearly position Knit Thinking. Wilson (1998) and Ingold (2013) both discuss the importance of tactility and proprioception in knowledge generation through an anthropological/medical lens. Cross (2011), Dorst (2003), and Walker (2006), key proponents of Design Thinking, recognise the active reciprocity of designing and learning, placing the user at the core of design decisions. Craft Thinking, developed conceptually by, among others, Frayling (2001) and Adamson (2007), foregrounds materials, processes and skills in knowledge generation. Hemmings (2012), Nimkulrat et al (2016), and Igoe (2021) distil the knowledge embedded in the ubiquity of textile making into viable design research theory. All stress the importance of "the intelligence of the hand" Auch (2016).

Successful knit design requires a deep knowledge of materials, fabric structure and technology that can be applied creatively, creating a tension between technical limitations and creative outputs. Reya: Reuse Yarn addresses waste in high-volume knitwear manufacturing. The product it addresses is a knitted garment that foregrounds the needs of the consumer and positions high-volume knitwear production in the realm of an industry-based Design Thinking output. The process used to create the products in this research is materials/ process-based and is situated within Craft Thinking. Knit Thinking straddles pragmatic design solutions and textile-making theory, critically extending knowledge generation beyond the textile boundaries of product and process to wider societal practices. With consideration of the wider societal problems the research addresses, the research question was refined further:

“How can the knowledge embedded in the research outputs be revealed to address pre-consumer waste in commercial knitwear production, and how can the findings be applied more widely?”

5. Methods

To date, Reya: Reuse Yarn has three development and testing stages. The Pilot translated the speculative insubstantiality of colour research to a pragmatic design solution and demonstrated that the research method was appropriate to the research question. Commercial Trial 1 (CT1) tested the research as a product with consumers. Commercial Trial 2 (CT2) tested the design process, supported by an Educational Resource (not discussed in this paper).

To utilise deadstock yarn in commercial production successfully, any new technique development must overcome the three main technical barriers to commercial adoption from the high-volume fashion knitwear industry: aesthetics/newness, quantity of materials available and changes to fabric properties with yarn substitution. Producing garments in fabrics that demonstrated optical mixing was evaluated as a method for fulfilling the three required conditions.

5.1 The Pilot: translation of theory to product/ process

The pilot used hand-knit production and produced fabrics equivalent to 5gg knitwear. The hand-knit process was an important element of the craft methodology, enabling a close relationship between maker and materials (Brinck, 2024) in the symbiosis of material properties, fabric construction and colour perception through 3D sketching.

5.1.1 Aesthetic newness

Prior research had utilised 2-colour and 3-colour birds-eye jacquard structures to promote conditions for optical mixing. The optically mixed colour was perceived as different from the physical colours that produced it, thus creating a new fabric aesthetic. The fabrics had been produced using solid, highly saturated, coloured yarns and successfully demonstrated optical mixing, particularly at a distance, which enhances the effect (Gage 1993). To increase the visual effect at the closer range of a garment required a variation of colour saturation and colour area.

Lambert et al (1986) demonstrated the increased effect of optical mixing as the intimacy of fibre blends in yarns increases, from solid dyed to marl and melange effects. As this research used pre-spun yarns, a blended effect was created by bundling three ends of different coloured or textured yarns together. Using multi-ends of yarn is a common practice in commercial knitwear production when using lower value yarns that lack character, for example, 4-5 ends of a smooth 2/30Nm acrylic may be used in 5gg knitwear production. The loose bundling adds loft to the yarn and produces a round profile that enhances stitch definition. The three ends ran parallel to each other; this was not controlled during the knit process, therefore, the colour on the face of each stitch was random, producing a micro-stripe within the fabric. Often, each stitch displayed all three contributing colours, significantly reducing the visible area of each colour. Uniformity could be achieved by twisting yarns together before knitting, although many knitwear manufacturers do not have this facility.

Optical mixing works across the colour gamut but a red, yellow and blue base for each yarn was chosen as these maximised the visible differences between yarns. Desaturated versions of each colour story were used. Swirnoff (2003) identified that decreasing the saturation of a colour increases its

propensity to perceptual change. In addition to a perceptual change in hue, optical mixing further desaturates colour and reduces colour value (lightness); it therefore influences all of the three colour characteristics and maximises colour newness. Yarns of varying hairiness/texture were used as texture reduces border contrast and therefore increases optical mixing. The blended yarns were produced with a tacit understanding of material properties suitable for knitwear.

5.1.2 Yarn Quantity

The test fabric was a birds-eye, float jacquard. Three blended yarns were used overall but only two were used in each course; these followed a clear rotation of A+B, C+A, B+C (figure 3, left and centre). Each blended yarn was produced from three individual deadstock yarns; the finished fabrics used nine yarns overall. Therefore, the weight of each yarn used is 1/9th of the total fabric weight, enabling smaller quantities to be used.

Lead Feeder	Trail Feeder
Colour B	Colour C
Colour C	Colour A
Colour A	Colour B





Figure 3 Left: Yarn order for birds-eye jacquard. Figure 3 Centre: Fabric detail. Figure 3 Right: Look Closer, See the Extraordinary, Pannett Art Gallery.

5.1.3 Fabric properties

The construction method addressed the first two barriers to a viable use of deadstock yarn in large-volume production; however, as the quantities of deadstock yarn available are variable, it was presumed that to fully utilise the blending method, some yarn substitutions would be required. This is known to change both fabric colour and fabric handle. The pilot not only tested the production method but also the extent to which changing one yarn within a blend affected the overall fabric colour or handle.

The pilot study, part of a Royal Society funded project at the Pannett Art Gallery, Whitby, North Yorkshire (Gaston, 2024), produced 16 hand-knitted, half-scale garments (figure 3, right). The three colour blends produced for each garment demonstrated a wide variety of fibre, colour hue, value and saturation, yarn count and texture. However, whilst not identical, the resultant fabrics demonstrated remarkable similarities in both colour and texture. In an uncontrolled survey, 81% of respondents (n=311 exhibition visitors) matched more than one version of the physical garments displayed to a photographic image of one of the garments, presented in proximity. This suggests that accurate colour reproduction was less important to the participants than the complete garment aesthetic. This is the converse of all existing industry-focused colour research for fashion, which focuses on precise repeatability of colouration and accurate colour communication (Bozzi et al, 2023). As the participants

did not have the opportunity to touch the garments, the fabric handle was not assessed. However, none of the participants identified the visual effect of textural differences in their feedback. Rubin (1921) found that in visual perception, overall form had primacy over border texture. This could explain the lack of attention given to the surface texture of the garments that were all the same design.

Throughout the production of the mini-garment collection, intuitive material use was refined by what Brinck (2024) describes as a coupling between materials and maker, an understanding that develops through expert making. The pilot clearly demonstrated the potential for integrating deadstock yarn use in commercial manufacturing. However, the Knit Thinking methodology extended this from a product or process solution to an active step towards a new manufacturing practice if all stakeholders can accept minor colour inconsistency within one garment style.

5.2. Commercial Trial 1: Product, Process and the Consumer.

In the consumer-led fashion industry, manufacturers will not accept new colouration processes without evidence of consumer acceptance. Contemporary colour research focuses on accuracy and repeatability with accepted tolerances of $\Delta E<1.5$ or *one shade lighter or darker* (Samanta, 2022). However, there isn't a baseline for consumer attitudes towards the acceptance of colour differences in individual garments. CT1 was designed to provide an initial understanding of acceptable colour variation tolerances from a consumer perspective. It was also an opportunity to refine the production process. As an industry-facing solution, it was important to follow conventional manufacturing processes as closely as possible, whilst maintaining the three-colour float jacquard fabric structure using blended yarns.

The trial addressed the low quantities of deadstock yarns available and aimed to test whether replacing yarns with similar but not identical yarns affected garment aesthetics or fabric handle. Small quantities of deadstock yarns (0.5kg-3.5kg/colour) were provided by project partners John Smedley. The colour range of the donated yarns was random, but it was possible to produce colour blends that contained yellow, red and blue, following the method used in the pilot study.

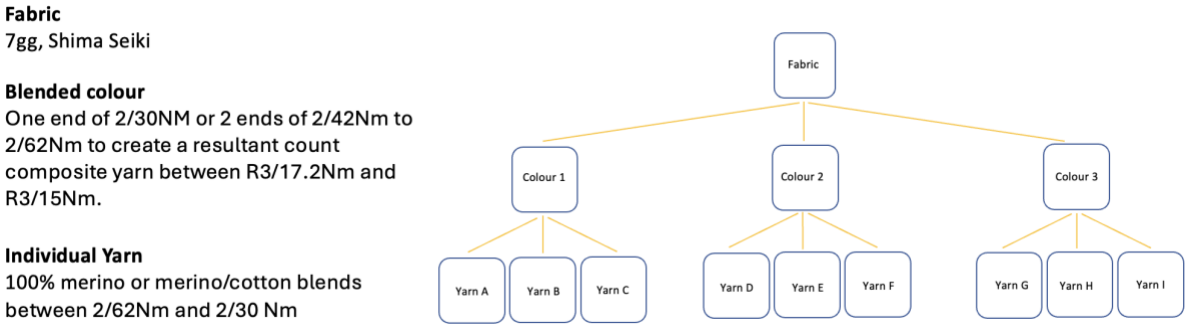


Figure 4: The hierarchy of yarns in blends and fabrics.

The blended design technique was used to produce ten crew neck, raglan sweaters (figure 5). During production, each yarn in a blend was replaced, sequentially, with one of similar, but not identical, colour and/or material properties. This resulted in ten garments with subtle, visual differences. Garments one and ten were produced from completely different yarn sets. Consistent use of trim colour reduced the design variables.



Figure 5: Ten garments with subtle yarn substitutions. Garment 1 (far left) and Garment 10 (far right) are completely different yarn blends.

Feedback was collected from 26 participants during five focus groups (Newcastle, Yorkshire x 2, London x 2). The participants, recruited from personal and professional networks by self-selection following a social media call, identified as either fashion professionals, design professionals, sustainability interest or general population. The research question positioned all respondents as consumers. Age and gender characteristics were not collected; however, the predominantly female participants ranged from young adults to retirees.



Figure 6: The mock online store used to test consumer reaction to minor colour inconsistency.

The focus group research question was posed as a simulation of an online purchase (figure 6). Participants were shown a page from a mock online store featuring one of the research garments, on a laptop. This was removed, and the participants were shown the ten physical garments as if they had

arrived in the post after purchase. The viewing conditions were not controlled, replicating domestic use. All focus groups were held in naturally well-lit rooms with additional, non-standard illumination. Whilst standard observer positioning was not employed, the size of each individual colour field (and therefore the effect of optical mixing) is affected by viewing distance, so to maintain consistency, the on-screen image was viewed from approximately 45cm by each participant, on a 13" lap-top placed directly in front of them. They were asked to select which of the garments they would accept if they had ordered the online version. The research task was followed by a general discussion facilitated through rose, bud, thorn analysis (Martin & Hannigan, 2012). The conversation was digitally recorded, and a transcript was thematically coded, from which findings were drawn.

Findings

CT1 produced knowledge about consumer attitudes to the product from quantitative assessment (figure 7) and discussion with participants in the research, and knowledge about the design process, revealed through the production of the garment collection. The findings were consistent, with no statistical variance across the participant groups.

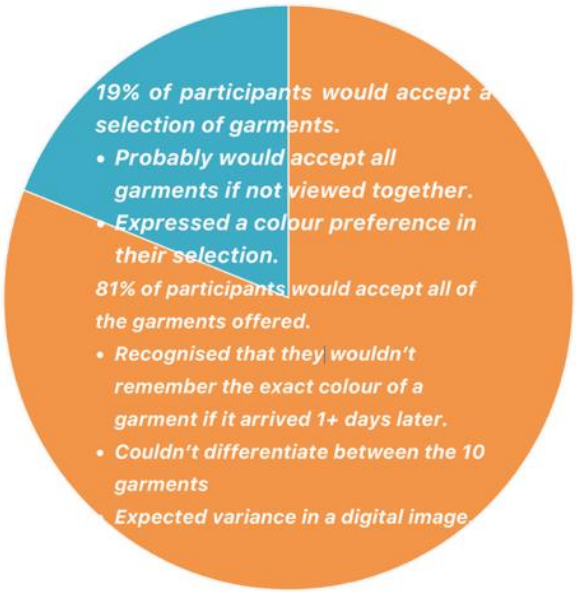


Figure 7: The quantitative findings from commercial trial 1.

Rather than viewing minor colour and material inconsistencies as a barrier to purchase, consumers from all the research categories welcomed the visual individuality resulting from using blended yarns in garment manufacture. This was recognised as a tangible benefit in a fast fashion market where consumers valued the possibility of purchasing a unique garment without additional cost. Additionally, they considered the reuse of deadstock yarn provided a clear, green narrative that could be used in a marketing strategy.

The act of producing physical garment outcomes revealed new process knowledge.

- When using a large colour gamut in one garment, mixing colour in the blended yarns in addition to the fabric structure produced more successful optical mixing.
- When similar colours were blended in each yarn, the fabrics appeared striped.

- Optical mixing reduced the colour intensity of each colour. This allowed fashion colours to be absorbed into a classic palette, potentially offering longevity.
- Individual yarns with highly saturated colour affected the overall colour perception of the garment, particularly in the red/orange hues.
- Participants regularly suggested that the method was particularly suitable for menswear or childrenswear, offering manufacturers the opportunity to pivot yarns to different markets.

Whilst CT1 employed a user-oriented Design Thinking methodology that separated production from materials (Kimbell, 2021), it extended the research in terms of Knit Thinking through the demonstrable acceptance of colour inconsistency by consumers. This is contrary to the accepted fashion industry position that the consumer demands accurate colour communication when making purchasing decisions. Similarly, it rejects the primacy of the retail industry position, which demands colour consistency. The trial was limited in its size and number of designs and materials tested. A more controlled trial, using standard observer conditions could be conducted to find absolute tolerances for consumer acceptance of colour variance.

5.3. Commercial Trial 2: Product, Process and the Designer

CT1 was limited to one aesthetic. To gain acceptance from the design community, CT2 sought to expand the aesthetic/design possibilities of yarn reuse through colour blending and produce this in a format that is demonstrable to the knitwear industry. Initial fabric experimentation was conducted by Dr Julia Wilmott, whose prior research investigated blending through pattern for improved fabric handle when using British wool (Wilmott, 2025). She utilised this knowledge to produce 40 fabrics, demonstrating the blending technique in a range of fabric structures.

This was applied to six garments, again produced on Shima Seiki CNC knitting machines using John Smedley deadstock merino wool yarn, in a count range 2/62Nm -2/30Nm. Each garment was also a colour experiment that either tested the ability of a blend to reduce the impact of substitutions in the content of the blend (1 - 4), or to test the aesthetic effects of different colour distribution in blends (5 & 6) (figure 7).



1, A single blend containing one bright colour in a multi-end blend of desaturated and classic colours.



2, Disguising hue variation through contrast.



3, A comparison of a blend made from very similar hues.



4, Using the same neutral colour in each blended yarn to reduce impact of change

5, A comparison of blends that distribute tonal value with blends with distinct tonal values.

6, A comparison of blends that distribute hues through the blends with blends with distinct hues.

Figure 7: The six colour experiments as garments.

Garments in CT1 were produced in high-quality merino wool with excellent fabric handle; the feel of the garments was mentioned several times. This could have affected consumer perceptions, and so five garments were additionally knitted in low-cost acrylic yarn, reflecting yarn use in the fast fashion industry. These garments demonstrated transition through a broad tonal range, reflecting the methodology used in the prior research.

***Key findings*:**

- All colour experiments demonstrated successful blending in a range of aesthetics.
- Early analysis suggests that disguising trend-led colour in multicolour blends and disguising hue variation through contrast are particularly useful methods when using *difficult*, trend-led colours.
- Creating marl effects with very similar hues is particularly useful in mitigating minor seasonal updates when using classic colours such as neutrals or navy blue.
- The use of acrylic yarns did not diminish the acceptance of minor colour inconsistencies demonstrated in high-quality wool yarns, confirming the success of the technique as opposed to consumer preference for material use.
- Transitioning to using a blended colour approach is easily achievable and depending on the styles chosen to produce in blended coloured yarn, reprogramming of machine specifications may not be necessary.
- The effect of changing fibre was inconclusive in that 4 garments retained their original specifications, but one fabric structure (using tuck stitches) resulted in a smaller garment.

To date, testing of the CT2 garments has been less rigorous than that of CT1, but feedback collected during their exhibition at Design North, Northumbria University (3rd – 22nd

November 2025), included “totally inspiring”, “fabulous” and “the best exhibition I’ve seen at the University”. Feedback from industry visitors was equally positive and the financial benefit to industry was particularly acknowledged. Further industry feedback is ongoing.

6. Conclusion: Product/Process/Practice

Reya: Reuse Yarn explored the reduction of deadstock yarn in the large volume knit industry through the three domains of knit; product, process and practice. By addressing the barriers (figure 8) to using deadstock yarn in knitwear production, the tangible outcome of this research is a potentially viable commercial product utilising material innovation and the process to enable the use of deadstock yarn in high-volume knitwear production. The process used blending to produce new aesthetics in fabric outcomes for the yarn. Blending yarns reduced the weight of individual yarns required in each garment allowing larger production runs. Substituting similar yarns had little impact on overall handle and sizing but further work is needed when substituting yarns with different fibre content. Whilst most strategies for material reuse in the fashion industry are not yet scalable, this low-tech method is available using common manufacturing processes and could be implemented immediately with no additional cost. Converse to other material-led strategies, Reya: Reuse Yarn could have the greatest impact in fast fashion markets, where low-value waste yarn is routinely destroyed.

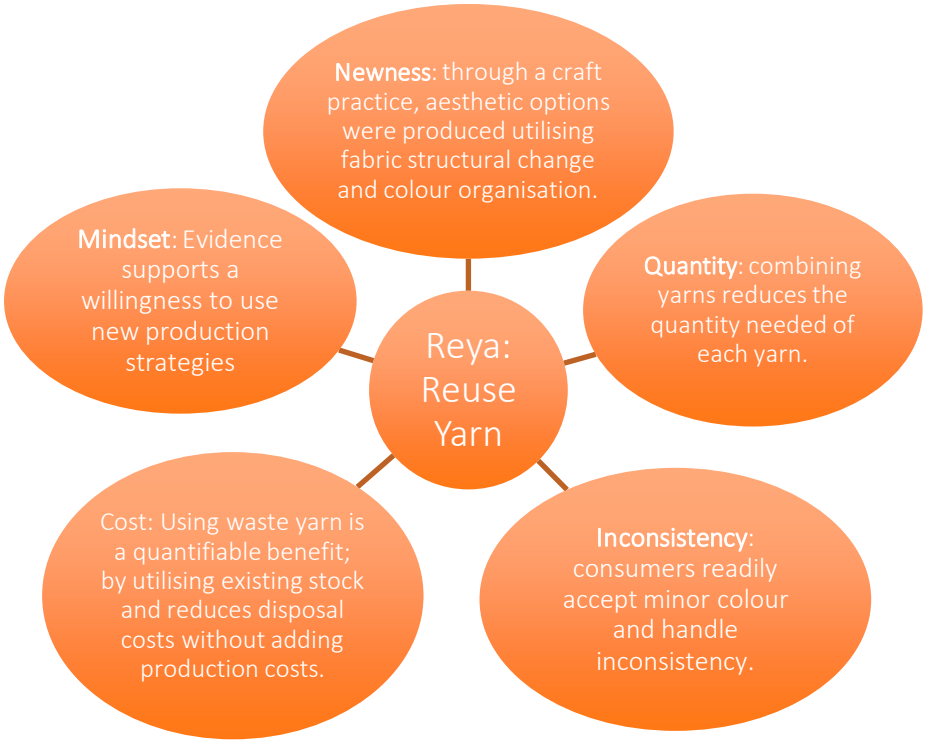


Figure 8: The approaches taken to overcome barriers to the reuse of deadstock yarn in high-volume knitwear production.

As a practice, acceptance of colour inconsistency is contrary to all conventional colour research, which foregrounds accurate repeatability and colour communication. Using the interdisciplinarity of knitted textiles and visual colour psychology as a provocation to challenge established norms in fashion manufacturing, Knit Thinking was used to contest the assumption that consumers always demand accurate colour matching. Whilst repeatability and accuracy are important in many cases, increasing acceptable variance levels to utilise deadstock will prevent waste. Similarly, a designer’s colour choice is very specific, leading to the rejection of the use of stock colours or minor dye-house inaccuracies.

Greater tolerance to colour variance could encourage the use of stock yarns, therefore minimising over-purchase. The evidence generated by Reya: Reuse Yarn has the potential to change this embedded view and produce a change to manufacturing mindsets. This will lead to positive environmental change and pre-empting Extended Producer Responsibility (EPR) Legislation. Challenging accepted norms across manufacturing could enable the development of new, more sustainable solutions within fashion and beyond. However, a knowledge gap still exists. This has been tested in small-scale production. It is now necessary to test at a full commercial scale. The next steps will be a manufacturer collaboration and supporting design guidelines. Additionally, testing colour variance systematically will enable the development of a baseline of tolerance to ΔE variance.

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